SIXTH SEMESTER B.A./B.Sc. DEGREE EXAMINATION, MARCH 2020

(CUCBCSS—UG)

Physics/Applied Physics

PHY 6B 11/APY 6B 12—SOLID STATE PHYSICS, SPECTROSCOPY AND LASER PHYSICS

Time : Three Hours Maximum : 80 Marks

Section A

Answer in a word **or** *phrase.*

Answer **all** *questions ; each question carries* 1 *mark.*

1. Expand SQUID.
2. Atomic packing factor of a diamond cubic structure is ————.
3. Give an example for simple cubic structure.
4. Which theory explains the superconductivity ?
5. Transition temperature of mercury is ————.
6. The temperature at which the band gap disappears in the case of superconductor is ————.
7. For a non rigid rotator the spacing between the successive spectral lines decreases, true or false
8. Population inversion is a phenomenon in ————.
9. Laser light is highly monochromatic, true or false
10. A superconductor exhibits complete Meissner effect is called ————.

(10 × 1 = 10 marks)

Section B

Answer in a short paragraph-three **or** *four sentences.*

Answer **all** *questions ; each question* carries 2 *marks.*

11. What is two fluid model ?
12. State Silsbee's Rule.
13. What is cooper pair ?

14. What is mean by coherence length ?

15. Differentiate spontaneous and stimulated emission

16. Define critical magnetic field.

17. What is zero point energy ?

(7 × 2 = 14 marks)

Section C

Answer in **a paragraph** *of about* **half a page to one page.**
Answer any **five** *questions ; each question carries* 4 *marks.*

18. Explain how X- rays are used for determining the crystal structure.

19. Define the terms : lattice, basis, crystal structure and unit cell.

20. Show that the packing factor of bcc lattice is $\pi\sqrt{3}/8$.

21. Discuss medical applications of laser ?

22. Give the salient features of BCS theory of superconductors.

23. Explain Raman effect.

24. What are Einstein's co-efficients ?

(5 × 4 = 20 marks)

Section D

(Problems · write all relevant formulas. All important steps carry separate marks)
Answer any **four** *questions ; each question carries* 4 *marks.*

25. In the microwave spectrum of identical rigid diatomic molecules, the separation between the spectral lines is recorded to be 0.7143 cm^{-1}. Find the moment of inertia of the molecule.

26. Calculate the inter planar spacing for (321) plane in simple cubic lattice with inter atomic spacing $a = 4.21\ A^0$.

27. Calculate the critical current which can flow through a long thin superconducting wire of A1 of diameter 10^{-3} m. The critical magnetic field for A1 is 7.9×10^3A/m.

28. Derive the expression for volume of unit cell in body centred cubic structure.

29. Derive London's equations and explain how it solution explains Meissner effect.

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30. The critical temperature, Tc for mercury with isotopic mass 199.5 is 4.185 K. Calculate its critical temperature when its isotopic mass changes to 203.4.

31. Calculate the angle of incidence at which electrons of energy 100 eV must be incident on the lattice planes of a metal crystal in order to give a strong Bragg reflection in the first order, given that the lattice spacing is 0.252 nm.

(4 × 4 = 16 marks)

Section E

Essays - Answer in about **two pages.**

Answer any **two** *questions ; each question carries* 10 *marks.*

32. Explain Bragg's law and Bragg's X-ray spectrometer.

33. Explain the theory and working of semiconductor laser.

34. Explain the crystal system in detail with examples.

35. Describe an experimental arrangement to study Raman effect. How is Raman scattering different from Compton scattering ?

(2 × 10 = 20 marks)